

Project 4 Phase 1: Deep and Un-Deep Visual Inertial Odometry

Filippo Marcantoni
Robotics Engineering, WPI
fmarcantoni@wpi.edu

Prahladh Raja
Robotics Engineering, WPI
pnamboorkrishnam@wpi.edu

Abstract—This report describes the Phase 1 implementation of a stereo visual-inertial odometry pipeline based on the Multi-State Constraint Kalman Filter (MSCKF). The work was carried out on top of the provided starter code, and modifications were limited to the estimator functions specified by the assignment. The completed system initializes gravity and gyroscope bias from a short stationary segment, propagates the IMU state and covariance with incoming inertial measurements, augments the filter with cloned camera poses on each image, stores stereo feature observations across the sliding window, and performs Extended Kalman Filter (EKF) measurement updates using multi-view MSCKF constraints. The estimator is evaluated on the EuRoC MH_01_easy sequence against Vicon ground truth. Following the assignment specification, the estimate is aligned to the ground truth with an SE(3) transformation and evaluated with the RPG trajectory evaluation toolbox.

I. INTRODUCTION

Visual-inertial odometry (VIO) estimates the motion of a moving platform by combining images with inertial measurements. The two modalities are complementary. The IMU reports high-rate angular velocity and linear acceleration, which are reliable during fast motion and in the gaps between camera frames, but pure inertial integration drifts quickly because of sensor noise and bias. Cameras provide geometric constraints from tracked features that bound this drift, but feature tracking can fail under motion blur, changing illumination, or low texture. A good VIO system therefore has to fuse both sources so that the IMU handles short-term accuracy while vision controls long-term drift.

The starter code for Phase 1 already provides the frontend and the overall framework of a stereo MSCKF estimator, following the structure of the open-source `msckf_vio` implementation [4], but several core backend routines are left blank. The goal of this phase is to complete these routines and evaluate the resulting system on the EuRoC Machine Hall sequence MH_01_easy. The implementation follows the classical MSCKF formulation of Mourikis and Roumeliotis [1], together with the stereo extension by Sun *et al.* [2]. Because the assignment focuses on completing the starter code rather than redesigning the estimator, all changes were kept inside the requested functions and the evaluation was carried out using the RPG toolbox [5] to produce the plots and metrics.

The required Phase 1 deliverables are a screen capture of the live visualization, trajectory plots comparing the Vicon ground truth and the estimate, and the RMSE ATE after an

SE(3) alignment. The final run produces the screen recording `Output.mp4`; all trajectory and error figures reported in this report are generated by the RPG trajectory evaluation toolbox from the stamped estimate and ground-truth files.

II. DATASET

The system is evaluated on the Machine Hall 01 easy (MH_01_easy) subset of the EuRoC micro aerial vehicle dataset [3]. The sequence was recorded on a quadrotor carrying a forward-facing global-shutter stereo camera and a time-synchronized IMU. A Vicon motion-capture system provides six-degree-of-freedom ground-truth pose, which is used as the reference trajectory for all quantitative evaluation. MH_01_easy has moderate motion and good scene texture, so it is well suited for validating the baseline behavior of a newly implemented estimator.

III. METHOD

A. State Definition

The estimator uses the standard MSCKF nominal IMU state

$$\mathbf{x}_I = [{}^G_I\mathbf{q} \quad \mathbf{b}_g \quad {}^G\mathbf{v}_I \quad \mathbf{b}_a \quad {}^G\mathbf{p}_I \quad {}^I_C\mathbf{q} \quad {}^I\mathbf{p}_C]^T, \quad (1)$$

where ${}^G_I\mathbf{q}$ is the IMU orientation, $\mathbf{b}_g$ and $\mathbf{b}_a$ are the gyroscope and accelerometer biases, ${}^G\mathbf{v}_I$ and ${}^G\mathbf{p}_I$ are the IMU velocity and position in the global frame, and the last two terms describe the camera–IMU extrinsics. The full filter state also holds a sliding window of cloned camera poses. These clones let features tracked across several images constrain the trajectory without adding the corresponding 3D landmarks to the state, which is the defining feature of the MSCKF.

The raw IMU readings are bias-corrected as

$$\boldsymbol{\omega} = \boldsymbol{\omega}_m - \mathbf{b}_g, \quad \mathbf{a} = \mathbf{a}_m - \mathbf{b}_a, \quad (2)$$

and the continuous-time nominal kinematics are

$$\dot{\mathbf{q}} = \frac{1}{2}\boldsymbol{\Omega}(\boldsymbol{\omega})\mathbf{q}, \quad \dot{\mathbf{v}} = \mathbf{R}(\mathbf{q})^T\mathbf{a} + \mathbf{g}, \quad \dot{\mathbf{p}} = \mathbf{v}, \quad (3)$$

with $\mathbf{g}$ the gravity vector in the global frame. The quaternion rate matrix used throughout is

$$\boldsymbol{\Omega}(\boldsymbol{\omega}) = \begin{bmatrix} -[\boldsymbol{\omega}]_{\times} & \boldsymbol{\omega} \\ -\boldsymbol{\omega}^T & 0 \end{bmatrix}. \quad (4)$$

B. Error-State Dynamics

The linearized error state evolves as

$$\dot{\delta \mathbf{x}} = \mathbf{F} \delta \mathbf{x} + \mathbf{G} \mathbf{n}, \quad (5)$$

where $\mathbf{F}$ is the continuous-time state-transition Jacobian and $\mathbf{G}$ maps gyroscope and accelerometer noise into the error state. The nonzero blocks of $\mathbf{F}$ used in the implementation are

$$\mathbf{F}_{\theta\theta} = -[\boldsymbol{\omega}]_{\times}, \quad \mathbf{F}_{\theta b_g} = -\mathbf{I}, \quad \mathbf{F}_{v\theta} = -\mathbf{R}^T[\mathbf{a}]_{\times}, \quad (6)$$

$$\mathbf{F}_{v b_a} = -\mathbf{R}^T, \quad \mathbf{F}_{pv} = \mathbf{I}. \quad (7)$$

C. Stereo Measurement Model

Each stereo observation is stored as

$$\mathbf{z}_j = [u_0 \quad v_0 \quad u_1 \quad v_1]^T, \quad (8)$$

with (u_0, v_0) and (u_1, v_1) the normalized coordinates in the left and right cameras. For a triangulated feature seen from several cloned camera poses, the residual is

$$\mathbf{r} = \mathbf{z} - \hat{\mathbf{z}}, \quad (9)$$

where $\hat{\mathbf{z}}$ is the predicted stereo projection. The stacked Jacobian depends on both the filter state and the feature position; the feature dependence is removed through the nullspace projection described in Section IV-G.

IV. IMPLEMENTATION

The implementation stays close to the starter code. All changes are contained in the functions listed below, which follow the organization of `msckf.py`.

A. `initialize_gravity_and_bias`

An IMU has six degrees of freedom: three accelerometer axes and three gyroscope axes. When the device is stationary, the accelerometer should read $[0, 0, -g]^T$ and the gyroscope should read zero, but in practice both sensors carry bias and noise that have to be handled before any integration takes place.

This routine uses a short stationary segment at the start of the sequence to initialize the gyroscope bias and the gravity-aligned reference frame. The bias is set to the mean angular velocity,

$$\hat{\mathbf{b}}_g = \frac{1}{N} \sum_{k=1}^N \boldsymbol{\omega}_{m,k}, \quad (10)$$

and the gravity magnitude is taken from the norm of the mean acceleration,

$$\bar{\mathbf{a}} = \frac{1}{N} \sum_{k=1}^N \mathbf{a}_{m,k}, \quad \mathbf{g} = [0 \quad 0 \quad -\|\bar{\mathbf{a}}\|]^T. \quad (11)$$

The initial orientation is obtained by rotating the measured gravity direction onto the global gravity direction using the two-vector alignment helper provided with the starter code. This step matters because an incorrect gravity vector biases both the early IMU integration and every visual update that follows.

B. `batch_imu_processing`

At each image timestamp, this routine consumes all IMU packets buffered up to that time. It walks through them in chronological order, calls the process model on each, advances the current IMU identifier, and then discards the consumed packets from the buffer. This guarantees that the IMU state and covariance are always propagated up to the image timestamp before a visual update is applied.

C. `process_model`

The process model advances the nominal IMU state and its covariance by one IMU sampling interval using the bias-corrected angular velocity and linear acceleration. With the $\mathbf{F}$ and $\mathbf{G}$ matrices from Section III-B, the discrete transition matrix is approximated by a third-order series,

$$\Phi \approx \mathbf{I} + \mathbf{F}\Delta t + \frac{1}{2}(\mathbf{F}\Delta t)^2 + \frac{1}{6}(\mathbf{F}\Delta t)^3, \quad (12)$$

and the covariance is propagated with

$$\mathbf{P}_{k+1|k} = \Phi \mathbf{P}_{k|k} \Phi^T + \mathbf{Q}_d, \quad (13)$$

where $\mathbf{Q}_d$ is the discrete-time process-noise covariance obtained from $\mathbf{G}$ and the continuous-time noise spectral densities.

D. `predict_new_state`

Orientation, velocity, and position are integrated forward with a fourth-order Runge–Kutta scheme, which cuts numerical error compared with a first-order update and keeps the predicted motion stable between visual corrections. Writing the IMU kinematics as $f(t, \mathbf{y})$, the RK4 stages are

$$\mathbf{k}_1 = f(t_n, \mathbf{y}_n), \quad (14)$$

$$\mathbf{k}_2 = f(t_n + \frac{\Delta t}{2}, \mathbf{y}_n + \frac{\Delta t}{2} \mathbf{k}_1), \quad (15)$$

$$\mathbf{k}_3 = f(t_n + \frac{\Delta t}{2}, \mathbf{y}_n + \frac{\Delta t}{2} \mathbf{k}_2), \quad (16)$$

$$\mathbf{k}_4 = f(t_n + \Delta t, \mathbf{y}_n + \Delta t \mathbf{k}_3), \quad (17)$$

and the integrated orientation increment is projected back onto the unit quaternion manifold. The updated velocity and position are then written back to the current IMU state.

E. `state_augmentation`

When a new image arrives, the current IMU pose is cloned into the camera-state sliding window. The camera pose is computed from the current IMU pose and the known camera–IMU extrinsics:

$$\mathbf{R}_w^c = \mathbf{R}_i^c \mathbf{R}_w^i, \quad {}^G \mathbf{p}_C = {}^G \mathbf{p}_I + (\mathbf{R}_w^i)^T {}^I \mathbf{p}_C. \quad (18)$$

The covariance is augmented using the clone Jacobian

$$\mathbf{J} = \begin{bmatrix} \mathbf{R}_i^c & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} \\ [(\mathbf{R}_w^i)^T \mathbf{p}_C]_{\times} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{I} \end{bmatrix}, \quad (19)$$

so that the augmented covariance is

$$\mathbf{P}_{\text{aug}} = \begin{bmatrix} \mathbf{P} & \mathbf{P} \mathbf{J}^T \\ \mathbf{J} \mathbf{P} & \mathbf{J} \mathbf{P} \mathbf{J}^T \end{bmatrix}. \quad (20)$$

This step is what keeps the correlations between the IMU state and the cloned camera poses consistent, which later lets multi-view feature constraints flow back to the active inertial state.

F. add_feature_observations

On every image, this routine fetches the identifier of the active IMU state and updates the map server with the new stereo measurements. Features that are already registered have the current observation appended to their track under the active camera-state identifier, while previously unseen features are inserted as new tracks. The number of successfully tracked features is recorded at each step, and the tracking rate is computed as the ratio of tracked to available features. Keeping this bookkeeping correct matters because the MSCKF visual update is triggered only when a feature is lost or selected for pruning.

G. measurement_update

The visual update corrects the filter state by enforcing consistency between the predicted and observed stereo projections of features that have been tracked across multiple camera poses. The residual depends linearly on the error state,

$$\mathbf{r} = \mathbf{H} \tilde{\mathbf{x}} + \mathbf{n}, \quad (21)$$

with $\mathbf{H}$ the measurement Jacobian and $\mathbf{n}$ a zero-mean white noise term. To separate the state constraint from the triangulated feature position, the stacked Jacobian is projected onto the left nullspace of its feature-position columns:

$$\mathbf{H}_o = \mathbf{A}^T \mathbf{H}_x, \quad \mathbf{r}_o = \mathbf{A}^T \mathbf{r}. \quad (22)$$

When the resulting Jacobian is tall, a thin QR decomposition

$$\mathbf{H}_x = [\mathbf{Q}_1 \quad \mathbf{Q}_2] \begin{bmatrix} \mathbf{T}_h \\ \mathbf{0} \end{bmatrix} \quad (23)$$

is applied so that the full measurement information is preserved while the dimensionality of the update is reduced. The standard EKF equations follow,

$$\mathbf{S} = \mathbf{H} \mathbf{P} \mathbf{H}^T + \mathbf{R}, \quad \mathbf{K} = \mathbf{P} \mathbf{H}^T \mathbf{S}^{-1}, \quad \delta \mathbf{x} = \mathbf{K} \mathbf{r}, \quad (24)$$

and the correction is applied additively to the Euclidean components of the state and multiplicatively, through a small-angle quaternion product, to the orientation:

$$\mathbf{q}^+ = \delta \mathbf{q} \otimes \mathbf{q}^-. \quad (25)$$

Finally, the covariance is updated and explicitly symmetrized,

$$\mathbf{P}^+ = (\mathbf{I} - \mathbf{K} \mathbf{H}) \mathbf{P}^-. \quad (26)$$

H. Evaluation Artifacts

The quantitative evaluation is performed with the RPG trajectory evaluation toolbox using the stamped estimate and Vicor ground-truth files included in the Phase 1 output folder. The same output folder contains the RPG configuration file, the RPG-generated trajectory plots, and the RPG-generated absolute and relative error plots. The live viewer remains close to the starter behavior, and the required screen recording `Output.mp4` is obtained by capturing the application during a full run on MH_01_easy.

TABLE I
RPG TOOLBOX ABSOLUTE TRAJECTORY ERROR ON EUROC
MH_01_EASY AFTER SE(3) ALIGNMENT.

Metric	Value
ATE RMSE	0.1006 m
ATE mean	0.0905 m
ATE median	0.0864 m
ATE minimum	0.0022 m
ATE maximum	0.2540 m
ATE standard deviation	0.0438 m
Number of matched poses	2842

V. EXPERIMENTAL EVALUATION

A. Dataset and Evaluation Protocol

The completed estimator is evaluated on EuRoC MH_01_easy, with the Vicor pose stream used as ground truth. Since the estimator runs in its own local coordinate frame, a direct pointwise comparison is not meaningful before alignment. Following the assignment guidelines, the RPG trajectory evaluation toolbox is used to perform an SE(3) alignment and to compute the ATE. The toolbox configuration is

$$\text{align_type} = \text{se3}, \quad \text{align_num_frames} = -1, \quad (27)$$

which applies a full-trajectory SE(3) alignment over all matched poses.

The alignment minimizes the sum of squared positional residuals,

$$\min_{\mathbf{R}, \mathbf{t}} \sum_{k=1}^N \|\mathbf{R} \hat{\mathbf{p}}_k + \mathbf{t} - \mathbf{p}_k^{gt}\|^2, \quad \mathbf{R} \in SO(3), \quad (28)$$

and the translational absolute trajectory error is

$$e_k = \|\mathbf{R} \hat{\mathbf{p}}_k + \mathbf{t} - \mathbf{p}_k^{gt}\|. \quad (29)$$

The RMSE ATE is then

$$\text{ATE}_{\text{RMSE}} = \sqrt{\frac{1}{N} \sum_{k=1}^N e_k^2}. \quad (30)$$

B. Absolute Trajectory Error

Table I summarizes the ATE statistics reported by the RPG toolbox on the aligned trajectory. The translational RMSE ATE is 0.1006 m, reported as approximately 0.101 m, and this is the main quantitative metric for Phase 1. The mean and median errors stay below 10 cm, and the maximum per-pose error of 0.2540 m corresponds to localized degradation during the more dynamic parts of the sequence.

C. RPG Toolbox Plots

The top and side trajectory views generated by the RPG toolbox are shown in Fig. 1. These plots provide the trajectory comparison used in this report and show the aligned estimate against the Vicor reference. They also make it easier to tell apart horizontal-plane error from vertical-profile error, which

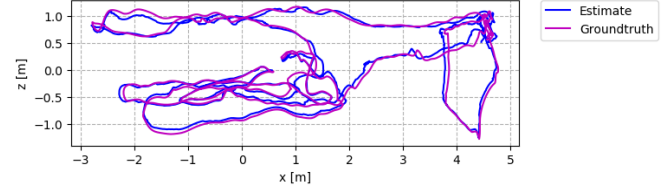
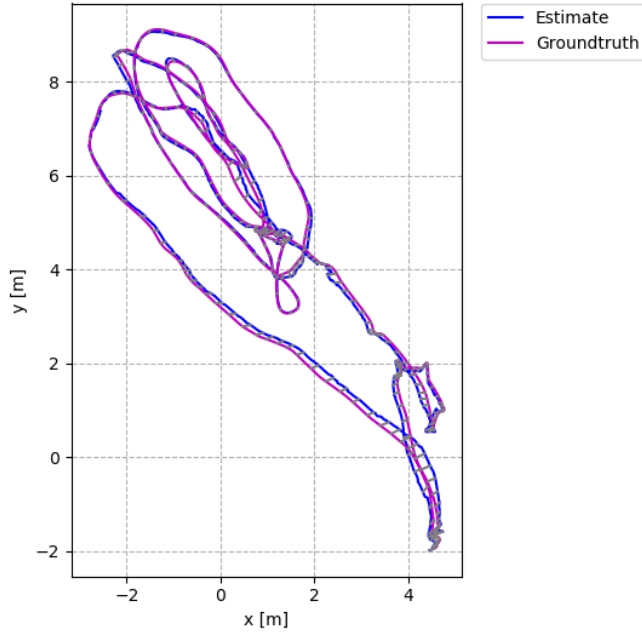


Fig. 1. RPG trajectory evaluation plots after SE(3) alignment. Left: top view. Right: side view.

helps in judging whether the estimator drifts more laterally or in altitude.

Figure 2 shows the absolute translation, rotation, and scale error plots produced by the RPG toolbox. The translation plot matches the ATE statistics in Table I directly, while the rotation and scale plots are included as supplemental diagnostics.

Figure 3 shows the relative error plots produced by the RPG toolbox, showing the measure drift over subtrajectory segments of varying length and therefore complement the global ATE metric. RMSE ATE is the main quantity requested by the assignment, while the relative plots are included here showing local consistency of the estimator.

VI. DISCUSSION

The main implementation challenge was staying consistent with the MSCKF formulation while changing as little of the starter code as possible. The estimator is sensitive to several details that can look minor in isolation: the gravity initialization has to produce a physically meaningful inertial frame, the process model has to propagate the covariance consistently with the nominal state, the state augmentation has to preserve the correlations between the IMU state and the cloned camera poses, and the measurement update has to inject visual constraints without adding landmarks to the state vector. Once these pieces line up, the estimator produces a smooth trajectory that tracks the Vicon reference closely over the full MH_01_easy sequence.

The second challenge was evaluation. The raw filter trajectory is expressed in a local frame that is not aligned with the Vicon frame, so a direct pointwise comparison is not meaningful before alignment. For that reason, an SE(3) alignment is applied before the ATE is computed. The reported

metric and all trajectory and error plots used in this report come straight from the RPG trajectory evaluation toolbox, which satisfies the assignment requirement and makes the result easier to reproduce.

The residual trajectory error is small in absolute terms but is not zero. The most likely contributors are feature-tracking outliers, linearization error in the filter, imperfect camera-IMU calibration and time synchronization, and short segments of the sequence where visual constraints are weaker. These are typical limitations of classical filter-based VIO systems and motivate the extensions described next.

VII. RESEARCH DIRECTIONS

A. Classical VIO

One natural direction for classical VIO is more robust feature handling. Better outlier rejection, adaptive feature selection, and stronger geometric consistency checks would likely improve performance under motion blur, repetitive texture, and poor feature distribution. Online spatial and temporal calibration is another important direction, since even small errors in camera-IMU extrinsics or timestamp alignment can hurt estimator consistency during aggressive motion. A third direction is to combine local filter-based VIO with loop closure and global pose-graph optimization, so that the efficient short-term estimation of MSCKF is paired with better long-term consistency on longer trajectories. Observability-aware filtering [2] remains a further key direction, because keeping the correct unobservable directions is essential for a stable and consistent visual-inertial estimator. Finally, an event-camera-based VIO system will be an interesting direction to pursue. Instead of standard frames, event cameras produce asynchronous brightness-change events with microsecond latency

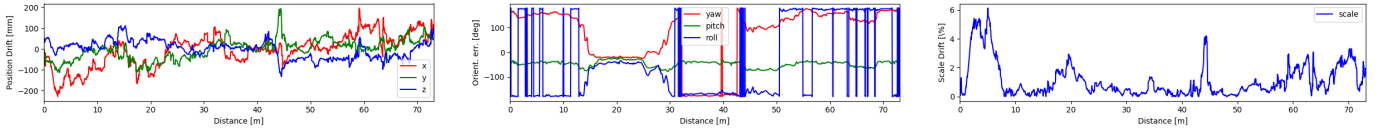


Fig. 2. Absolute error plots generated by the RPG trajectory evaluation toolbox. Left: translational ATE. Center: rotational error. Right: scale error diagnostic.

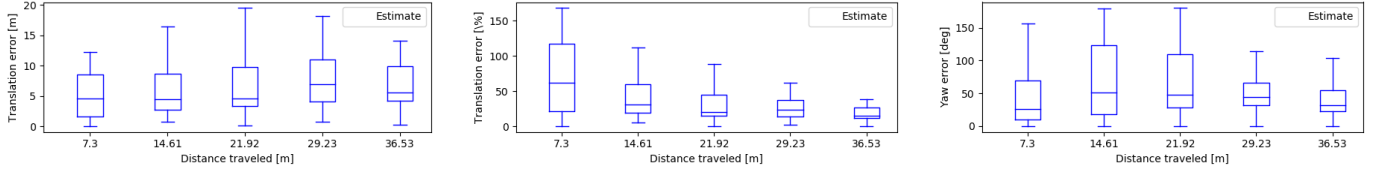


Fig. 3. Relative odometry error plots generated by the RPG trajectory evaluation toolbox. Left: relative translation error. Center: relative translation percentage error. Right: relative yaw error.

and high dynamic range. Combining IMU with event streams will be different because it works well in fast motion, low-latency, and difficult lighting conditions. The challenge is that the measurement model is completely different from standard reprojection residuals.

VIII. CONCLUSION

Phase 1 completes the stereo MSCKF backend system and evaluates the resulting estimator on EuRoC MH_01_easy. The completed system initializes gravity and gyroscope bias from a stationary segment, propagates the IMU state and covariance, augments the filter with cloned camera poses on each image, stores stereo feature observations across the sliding window, and applies multi-view MSCKF constraints through an EKF correction step. The live visualization was recorded as `Output.mp4`. For quantitative evaluation, the estimated trajectory was aligned to Vicon ground truth with an SE(3) transformation and assessed with the RPG trajectory evaluation toolbox. The resulting translational RMSE ATE is 0.1006 m, or approximately 0.101 m. The accompanying trajectory and error plots confirm that the estimate follows the Vicon reference closely, meeting all the requirements for Phase 1.

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